Qn. Set Code-7(SOT)

Semester: 5th
Programme: B.Tech
Branch: Comp. Engg.

AUTUMN END SEMESTER EXAMINATION-2025

5th Semester B.Tech

MACHINE LEARNING CONCEPTS

CS30037

(For 2022 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

(a) What is the specific name of the metric that represents the harmonic mean of a classifier's precision and recall?

(b) Which partitioning clustering algorithm is most sensitive to the initial random placement of cluster centers?

(c) Which issue in the data causes the Single Layer Perceptron to be unable to solve the XOR Problem?

(d) Which variant of Gradient Descent updates the model's parameters after processing each individual training instance?

(e) When a model performs exceptionally well on the training data but fails to generalize to new, unseen data, it is said to be suffering from what problem?

(f) What is the core naive assumption made by the Naive Bayes classifier about the relationship between features?

(g) Besides Entropy, name one other common impurity measure used to select the optimal split point in a classification Decision Tree.

(h) Define Mini-Batch Gradient Descent.

(i) Which linear dimensionality reduction technique seeks to find a lower-dimensional subspace that maximizes the variance of the projected data?

(j) In the context of a Hard Margin SVM, what are the critical data points that lie exactly on the boundary lines and dictate the position of the separating hyperplane?

SECTION-B

2. (a) Explain the relationship between Model Complexity and the Bias-Variance Tradeoff. Provide one specific technique to increase a model's complexity (to address Underfitting) and one technique to decrease it (to address Overfitting). [5]

(b) Justify why a decreasing learning rate is often more effective than a fixed learning rate for training deep neural networks using Gradient Descent variants. [5]

3. (a) Define the terms Core Point and Border Point in the DBSCAN clustering algorithm. Explain why DBSCAN is considered more robust to outliers than K-Means and discuss its capability to discover clusters of arbitrary shape. [5]

(b) Write down the cost function for both Lasso Regression and Ridge Regression. Justify why Lasso regression is preferred over Ridge regression. Derive the expression for computing the coefficients of Ridge Regression, formulate its pseudo code, and justify that it is a shrinkage method. [5]

4. (a) A customer dataset has 5 points p1(1,2), p2(2,3), p3(5,7), p4(6,8), p5(10,1). Initialize centroids as c1=p1 and c2=p2. Calculate the new centroid coordinates for c1 and c2 after the first iteration of the K-Means algorithm using Euclidean distance. [5]

(b) Differentiate between Pre-Pruning (e.g., stopping early based on max depth) and Post-Pruning (e.g., Reduced Error Pruning) in Decision Trees. Provide a justification for why Post-Pruning is generally more effective in balancing the Bias-Variance trade-off. [5]

5. (a) For a single-neuron network with a Sigmoid activation function, an input X=[0.5] and weight W=0.8, The target output Y=1.0 .Calculate the gradient of the Loss (using Mean Squared Error) with respect to the weight after one forward and backward pass, assuming the bias b=0. [5]

(b) Explain the concept of Support Vectors in a Support Vector Machine (SVM) classifier. Demonstrate how the optimization objective of an SVM, which seeks to maximize the margin is related to minimizing the weight vector's magnitude. [5]

6. (a) Given a dataset with a covariance matrix mat=[[4,1],[1,2]],calculate the eigenvalues and eigenvectors. Use the result to determine the first principal component vector and the corresponding variance captured by this component in Principal Component Analysis (PCA). [5]

(b) For a 2-D dataset, the optimal separating hyperplane is defined by the equation W*X+b=0.If the weight vector is W=[4,-3] and the bias b=5, what is the equation of the positive support vector hyperplane? Calculate the margin between the two class boundaries. [5]

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